The Open University

Faculty of Science, Technology, Engineering and Mathematics

M303 Further pure mathematics

M303

TMA 04 2025J

Mainly covers Book D *Metric spaces 1* **Cut-off date 3 March 2026**

You will find instructions for completing TMAs in the Assessment resources area of the M303 website. Please remind yourself of these instructions before beginning work on each TMA. You can submit each TMA electronically by using the University's online TMA/EMA service or by post together with completed TMA form (PT3). In both cases the University must receive it by the cut-off date given above unless you have agreed an extension in advance with your tutor.

Half of each TMA is formative and the other half summative. The formative questions are designed to *extend* your understanding of the topics, while the summative ones place more emphasis on *assessing* your understanding of the topics. Both types of question will be marked by your tutor, but (unlike the summative questions) marks for the formative questions do not count towards your final grade. You should submit your solutions to both sets of questions to your tutor by the cut-off date.

The marks allocated to each part of a question are indicated in the margin.

Although many of the questions on these assignments require numerical answers, the questions invariably carry *method marks*.

The words listed below, when used in questions, should be interpreted as indicated.

prove, show, explain, justify	Clear reasoning and explanation for all steps are called for.
determine, find, devise, calculate, compute	An indication of the method used and all working in arriving at an answer should be given.
deduce	Clear explanation of how one result follows from another is required.
solve	Working must be shown. A numerical answer alone is not sufficient.
evaluate, give, write down, list	Answer suffices. No explanation need be given.
hence	No marks will be awarded for any alternative method.

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PLAGIARISM WARNING – the use of assessment help services and websites

The **work that you submit for any assessment/examination on any module should be your own**. Submitting work produced by or with another person, or a web service or an automated system, **as if it is your own** is cheating. It is **strictly forbidden** by the University.

You should not:

- provide any assessment question to a website, online service, social media platform or any individual or organisation, as this is an infringement of copyright
- request answers or solutions to an assessment question on any website, via an online service or social media platform, or from any individual or organisation
- use an automated system (other than one prescribed by the module) to obtain answers or solutions to an assessment question and submit the output as your own work
- discuss examination questions with any other person, including your tutor.

The University actively monitors websites, online services and social media platforms for answers and solutions to assessment questions, and for assessment questions posted by students. Work submitted by students for assessment is also monitored for plagiarism.

A student who is found to have posted a question or answer to a website, online service or social media platform and/or to have used any resulting, or otherwise obtained, output as if it is their own work has committed a disciplinary offence under our Code of Practice for Student Discipline. **This means the academic reputation and integrity of the University has been undermined.**

The Open University's Academic Conduct Policy defines plagiarism in part as:

- using text obtained from assignment writing sites, organisations or private individuals
- obtaining work from other sources and submitting it as your own.

If it is found that you have used the services of a website, online service or social media platform, or that you have otherwise obtained the work you submit from another person, this is considered serious academic misconduct and you will be referred to the Central Disciplinary Committee for investigation.

This TMA is based on Book D (Metric spaces 1), except for the formative Question 8, which covers the start of Book E.

Part A is **summative**. These questions assess your knowledge of the module. You must not discuss these questions in the forums, and your tutor is not allowed to help you directly.

Part B is **formative**. You should choose (at most) two questions to submit to your tutor for marking, and we suggest that one of these is Question 8 since this will help you with the next TMA.

Doing the formative questions will help you to understand key concepts covered in the text. You are encouraged to discuss these questions in the module forums and with your tutor (who is allowed to help you).

Question 8 is designed to help you with TMA 05. Question 9 covers material that is more central to the module, while Questions 10 and 11 are designed to expand your understanding beyond the core material.

Your tutor will mark all of the summative answers that you submit and and the first two of any formative answers that you submit. If you have attempted at least one formative question then when your TMA is returned you will be sent worked solutions to all the formative questions. Note that only your marks for Part A count towards your overall assessment score.

Part A Summative

Question 1 – 4 marks

This M303 question tests your understanding of real sequences and is based on material in Chapter 13.

(a) (i) Give an example (x_n) of a strictly monotonic decreasing sequence of real numbers that is convergent to 1.

(ii) Give an example (y_n) of a strictly monotonic increasing sequence of real numbers that is convergent to -1.

In each case give the formula for the general term of your sequence. [*2*]

(b) Let (z_n) be the sequence given by

$$z_n = \begin{cases} x_n, & \text{if } n \text{ is odd,} \\ y_n, & \text{if } n \text{ is even.} \end{cases}$$

Determine whether (z_n) is a convergent sequence. [*2*]

Question 2 – 14 marks

This M303 question tests your understanding of the definition of a novel metric. It asks about material covered in Chapters 14 and 16.

A (non-vertical) line L in the plane is uniquely determined by its intersection with the y-axis, a, and the angle $\theta \in (-\pi/2, \pi/2)$ it makes with the x-axis,

$$L = L_{a,\theta} = \{(x, a + x\tan\theta) : x \in \mathbb{R}\}.$$

Let the set of all such lines in the plane be denoted by X, so

$$X = \{L_{a,\theta} : a \in \mathbb{R} \text{ and } \theta \in (-\pi/2, \pi/2)\}.$$

Define a distance function Δ on the set X by

$$\Delta(L_{a,\theta}, L_{b,\phi}) = |a - b| + |\tan\theta - \tan\phi| \quad \text{for } L_{a,\theta},\ L_{b,\phi} \in X.$$

(a) Calculate the distance between the lines $L_{0,\pi/4}$ and $L_{1,-\pi/4}$. [*2*]

(b) Prove that Δ is a metric on X by showing that

(i) Δ satisfies (M1). [*3*]

(ii) Δ satisfies (M2). [*1*]

(iii) Δ satisfies (M3). [*3*]

(c) Determine which lines are in $B_\Delta(L_{0,0}, 1)$. [*2*]

(d) Determine if the sequence of lines $(L_n)_{n=1}^{\infty}$ given by $L_n = L_{-1/n,\pi/2-1/n}$ is Δ-convergent. [*3*]

Question 3 – 9 marks

This M303 question tests your understanding of sequences of functions. It asks about material covered in Chapter 15.

Throughout this question we use the Euclidean metric for $\mathbb{R}$.

For $n \in \mathbb{N}$, define a function $f_n \colon \mathbb{R} \to \mathbb{R}$ by

$$f_n(x) = \begin{cases} \sin(\pi x), & \text{if } 2n \le x < (2n+1), \\ 0, & \text{otherwise.} \end{cases}$$

(We are working in radians.)

(a) Sketch, by hand or otherwise, the graphs of f_1, f_2 and f_3. [*3*]

(b) Determine whether the sequence (f_n) has a pointwise limit. [*3*]

(c) Determine whether the sequence (f_n) converges uniformly on $\mathbb{R}$. [*3*]

Question 4 – 4 marks

This M303 question tests your understanding of continuity for functions with domain $C[0,1]$. It asks about material covered in Chapter 15.

Show that the function $F \colon C[0,1] \to C[0,1]$ given by

$$F(f)(t) = \int_t^1 (1 - x^4) f(x)\,dx \text{ for each } f \in C[0,1] \text{ and } t \in [0,1]$$

is $(d_{\max}, d_{\max})$-continuous on $C[0,1]$. [*4*]

Question 5 – 9 marks

This M303 question tests your understanding of material discussed in Chapter 16.

We define three subsets of the plane (using set notation) as follows:

$$A = \{(x, y) \in \mathbb{R}^2 : 2|x| \leq 3\}, \quad B = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 4\}$$

and

$$C = A \cap B.$$

(a) Sketch, by hand or otherwise, the set C, taking care to distinguish between points that lie inside and outside the set. [*2*]

When sketching sets you should follow the conventions used in the module text by using dashed lines to indicate points of the boundary that are not included in the set and solid lines to indicate points that are included. The status of individual points can be indicated by using small circles to indicate points that are not included and small filled-in circles to indicate points that are included.

(b) (i) Determine whether $(\frac{3}{2}, \frac{\sqrt{7}}{2})$ is a closure point of C for the Euclidean metric. [*2*]

(ii) Write down in set notation the closure, interior and boundary of C for the Euclidean metric. [*3*]

(c) Write down the closure of C for the 'mixed' metric d given by

$$d((x_1, y_1), (x_2, y_2)) = |x_1 - x_2| + d_0(y_1, y_2).$$ [*2*]

(Here d_0 denotes the discrete metric on $\mathbb{R}$. You may assume that d defines a metric for $\mathbb{R}^2$.)

Question 6 – 9 marks

This M303 question tests your understanding of material discussed in Chapter 16.

Let

$$D = \left\{ x \in \mathbb{R} : x = \frac{m}{p} \text{ for some } m \in \mathbb{N} \text{ and } p \text{ prime} \right\}.$$

(a) Show that D is countable. [*3*]

(b) (i) Show that for each $y \geq 0$ and each $r > 0$, there is $x \in D$ with $y \leq x < y + r$. [*3*]

(ii) Hence, or otherwise, show that D is dense in $[0, \infty)$ for the Euclidean metric. [*3*]

Question 7 – 1 mark

This question can be done at any point after Week 14, during your study of Chapters 13–16.

Watch the recording of the *first* informal online lecture on Book D that will be available in Meeting Recordings (of the informal online lecture room) by the end of Week 14. Write down the date when you watched it. Your tutor will be able to verify that you have viewed the recording. (If you are not able to do this, contact your tutor for advice.) [*1*]

Part B Formative

Question 8 – 9 marks

In this M303 question you will practise working with ideals in rings that are extensions of degree two over the integers. This question concerns Chapter 17 of Book E, and relates to Question 1 in TMA 05.

Let R be the ring $\mathbb{Z}[\sqrt{-13}]$.

(a) Let A be the subset of R defined by

$$A = \{s + t\sqrt{-13} : s, t \in \mathbb{Z},\ s - t \equiv 0 \pmod{7}\}.$$

Show that A is an ideal in the ring R. *[3]*

(b) The subset

$$B = \{u + v\sqrt{-13} : u, v \in \mathbb{Z},\ u + v \equiv 0 \pmod{7}\}$$

is also an ideal in the ring R (you do not need to prove this fact). Show that $AB = \langle 7 \rangle$. *[6]*

Question 9 – 9 marks

In this M303 question we ask you to sketch particular balls and spheres for various metrics on the plane. You should be able to attempt this question after reading Chapter 14.

Sketch, by hand or otherwise, the sets

$$B_d((0,0),1), \quad B_d[(0,0),1] \quad \text{and} \quad S_d((0,0),1)$$

for the metric d on the plane when d is given by each of the following, for $(a_1,a_2),(b_1,b_2) \in \mathbb{R}^2$. (You may assume that each of these defines a metric.)

(a) $d((a_1,a_2),(b_1,b_2)) = 6|b_1 - a_1| + |b_2 - a_2|$ *[3]*

(b) $d((a_1,a_2),(b_1,b_2)) = (|b_1 - a_1|^4 + |b_2 - a_2|^4)^{1/4}$ *[3]*

(c) $d((a_1,a_2),(b_1,b_2)) = d_0(a_1,b_1) + 6|b_2 - a_2|$ *[3]*

(Recall that d_0 is the discrete metric.)

Question 10 – 16 marks

In this M303 question we introduce a metric for the rational numbers, known as the 5-adic metric*, that is very different from the usual one for $\mathbb{Q}$ induced by the Euclidean metric for $\mathbb{R}$. You should be able to do this question after reading Chapters 14–15.*

Using results from Book A, one can show that for each non-zero rational number x, there is a unique $m \in \mathbb{Z}$ such that

$$x = 5^m \frac{a}{b},$$

where $a \in \mathbb{Z} - \{0\}$, $b \in \mathbb{N}$, and 5 does not divide either of a and b (you do not need to give a proof of this). We use this fact to define a function $f_5 \colon \mathbb{Q} \to \mathbb{R}$ by

$$f_5(x) = \begin{cases} 5^{-m}, & \text{if } x = 5^m \dfrac{a}{b} \text{ and 5 does not divide either of } a \text{ and } b, \\ 0, & \text{if } x = 0. \end{cases}$$

Now define a distance function $d_5 \colon \mathbb{Q} \times \mathbb{Q} \to \mathbb{R}$ by

$$d_5(r, s) = f_5(r - s).$$

(a) Write down $f_5(1)$, $f_5(5)$ and $f_5(5^n)$ for $n \in \mathbb{N}$. [2]

(b) Show that f_5 has the following properties.

(i) For each $x \in \mathbb{Q}$, $f_5(x) \geq 0$, with equality if, and only if, $x = 0$. [2]

(ii) For each $x \in \mathbb{Q}$, $f_5(-x) = f_5(x)$. [2]

(iii) For $x, y \in \mathbb{Q}$, $f_5(x + y) \leq \max\{f_5(x), f_5(y)\}$. [6]

(c) Using the results of part (b), or otherwise, show that d_5 is a metric for $\mathbb{Q}$. [4]

Question 11 – 16 marks

In this M303 question we define a metric on a group. The question revises some techniques from Book B in addition to enabling you to practise some techniques from Book D.

Let G be a group, let $n \in \mathbb{N}$, and suppose that $H_0, H_1, H_2, \ldots, H_n$ are subgroups of G such that

$$\{e\} = H_0 \leq H_1 \leq \cdots \leq H_n = G, \quad \text{with } H_i \neq H_{i-1} \text{ for } i = 1, 2, \ldots, n.$$

We define a distance function $d_G \colon G \times G \to \mathbb{R}$ as follows:

$$d_G(g, h) = k, \quad \text{where } gh^{-1} \in H_k \text{ and } gh^{-1} \notin H_i \text{ for all } i < k.$$

(a) Prove that d_G is a metric for G by showing the following.

(i) d_G satisfies (M1). [2]

(ii) d_G satisfies (M2). [3]

(iii) d_G satisfies (M3). [6]

(b) Show that a sequence (g_i) of elements in G is d_G-convergent if, and only if, the sequence (g_i) is eventually constant. [5]